Breast Cancer

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Microbiome Signatures, Microbiome Medicine

Abstract. Microbiome signature of breast cancer: explore the unique signature and microbiome-targeted interventions (MBTIs) for the disease.

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Overview

Breast cancer is the most prevalent cancer among women globally, accounting for over 2.3 million cases annually and representing a leading cause of cancer mortality.[1] Traditionally linked to genetic predispositions and environmental exposures, emerging evidence highlights the microbiome as a critical and underappreciated factor influencing breast cancer progression, immune response, and treatment outcomes.[3][x] Central to its development are mutations in tumor suppressor genes BRCA1 and BRCA2, which impair DNA repair, promote genomic instability, and increase susceptibility, particularly to aggressive subtypes like triple-negative breast cancer.[4] Screening strategies such as mammography and breast self-examination play critical roles in early detection, yet awareness and participation remain insufficient in many populations. Emerging evidence highlights the microbiome's relevance in breast cancer, with dysbiosis in gut and oral microbiota implicated in modulating estrogen metabolism, inflammatory responses, and immune surveillance, thereby influencing tumorigenesis and treatment outcomes.[5][6][7] Understanding the microbiome signatures associated with breast cancer offers unprecedented opportunities for targeted therapies, preventive strategies, and personalized treatment in breast cancer management.

Associated Conditions

Diabetes and benign breast disease (BBD) are interconnected risk factors that influence breast cancer risk. Type 2 diabetes (T2D) increases the risk of breast cancer through mechanisms like hyperinsulinemia and the activation of the insulin-like growth factor (IGF) pathway, which promotes cell growth and inhibits apoptosis, leading to a higher likelihood of cancer development, particularly in estrogen receptor-positive breast cancer.[8]Type 1 diabetes (T1D) also elevates breast cancer risk, especially shortly after diagnosis.[9] Meanwhile, a history of BBD, especially conditions like fibroadenoma and lobular proliferation, is a known risk factor for breast cancer. Surgical treatment for certain benign conditions, such as fibroadenoma, has been shown to reduce future cancer risk.[10] These factors, along with potential microbiome disruptions, can worsen insulin resistance and influence immune function, further elevating breast cancer risk. The gut microbiome, in particular, has been shown to play a role in breast cancer development, with a reduction in short-chain fatty acid-producing bacteria like *Faecalibacterium prausnitzii*, which are associated with anti-

inflammatory effects and maintaining metabolic health.[11][x] Dysbiosis, or an imbalance in gut bacteria, may contribute to increased systemic inflammation and other metabolic conditions that heighten breast cancer susceptibility.[12] Thus, managing diabetes, monitoring BBD, and understanding the role of the microbiome are crucial for reducing breast cancer risk.

Causes

Breast cancer is thought to arise from a multifactorial interplay of genetic, environmental, and hormonal factors, with up to 70% of cases having unknown etiology. However, recent studies find that dysbiosis—characterized by a significant overgrowth of *Proteobacteria* and reduced diversity of beneficial taxa like *Bacteroidetes* and SCFA-producing bacteria like *Faecalibacterium prausnitzii*, plays a critical role by disrupting estrogen metabolism, promoting inflammation, and altering the tumor microenvironment in breast cancer.[13][x] Evidence suggests that frequent antibiotic use may contribute to breast cancer risk by promoting *Proteobacteria* blooms as well as *Escherichia coli* and *Klebsiella sp.*, which are enriched in the breast cancer patient, amplifying systemic inflammation and immune suppression.[14][x][15] Microbial dysbiosis, both local and systemic, has significant implications for breast cancer progression, with some literature even proposing that antibiotics may potentially play a causal role in the condition.[16][x]

Diagnosis

Currently, breast cancer diagnosis relies heavily on imaging (mammography, ultrasound) and histopathological confirmation. However, advancements in microbiome research suggest that microbial signatures, including shifts in gut and breast tissue microbiota, could serve as non-invasive diagnostic biomarkers. For instance, altered ratios of *Firmicutes* to *Bacteroidetes* in the gut and specific microbial markers in breast tissue (e.g., *Bacteroides fragilis*, *Methylobacterium radiotolerans*) offer promising diagnostic potential. Integrating microbiome analyses into routine diagnostics could improve early detection, especially for aggressive subtypes like triple-negative breast cancer. [17]

Primer

The interplay between the microbiome, immune modulation, and hormone metabolism highlights the microbiome's central role in breast cancer pathogenesis. Dysbiosis contributes to systemic inflammation, immune escape, and tumor growth. For example, reductions in SCFA production impair anti-inflammatory pathways, while overgrowth of specific taxa drives estrogen metabolism disruptions, fueling hormone receptor-positive cancers. Addressing these microbial imbalances through targeted interventions may enhance therapeutic outcomes. Additionally, the activity of beta-glucuronidase, an enzyme involved in estrogen conjugation, is linked to the microbial dysbiosis observed in women with a history of breast cancer. [18]

Metallomic Signatures

Metallomic signatures play an important role in understanding breast cancer, with recent research highlighting how heavy metals and metabolites contribute to the progression of the disease.[19] A growing body of evidence suggests that exposure to heavy metals, including copper (Cu), cadmium (Cd), and zinc (Zn), can significantly influence breast cancer development by altering cellular pathways, including reactive oxygen species (ROS) production.[20] These metals can induce cancer progression by mimicking or

disrupting essential metabolic pathways, influencing cell growth and survival. Studies have shown that increased concentrations of Cu and Cd, combined with lower levels of Zn, are associated with breast cancer patients compared to healthy controls.[21][22] Cadmium's estrogenic effects are implicated in promoting breast cancer cell proliferation, suggesting a potential role in hormone-sensitive breast cancer. This is because cadmium, a heavy metal, can mimic the effects of estrogen by binding to estrogen receptors and activating them.[23] This activation can lead to increased cell growth and proliferation, contributing to the development of hormone-sensitive breast cancer.

▶ **What are the metallomic signatures of Breast Cancer?**

MMAs

The Major Microbial Associations (MMAs) for breast cancer can be identified as taxa that are consistently reported as being significantly altered and biologically relevant to disease pathogenesis. The MMAs for breast cancer reflect key microbial patterns that are enriched or depleted in affected patients, revealing functional dysbiosis contributing to disease progression. Enriched microbes, such as *Proteobacteria* and *Bacteroides*, drive pro-inflammatory pathways and systemic effects, exacerbating cancer-related outcomes. Conversely, depleted microbes, including *Faecalibacterium prausnitzii* and *Clostridiales*, represent a loss of anti-inflammatory and protective functions critical for maintaining gut and systemic health.[37][x] Additionally, microbes like *Prevotella amnii* and *Lactobacillus vaginalis* highlight specific effects on estrogen metabolism and mucosal immunity, which are especially relevant in postmenopausal breast cancer. Understanding these microbial associations offers insights into potential microbiome-targeted interventions and biomarkers for breast cancer.

▶ **What are the Major Microbial Associations (MMAs) for breast cancer**

MMA	Status in Breast Cancer	Key Role/Impact
Proteobacteria	Enriched	Associated with pro-inflammatory effects and systemic immune modulation, contributing to dysbiosis.
Enterobacteriaceae (guild)	Enriched	A major family within Proteobacteria; significant pro-inflammatory effects and disease-associated dysbiosis.
Actinobacteria	Depleted	Protective role in health; reduction, especially in postmenopausal women, marks disease-associated dysbiosis.
Bacteroides (guild)	Enriched	Involved in lipid metabolism, immune modulation, and inflammation, contributing to systemic cancer effects.
Faecalibacterium prausnitzii	Depleted	Anti-inflammatory properties and role in gut homeostasis make it a marker of health.
Clostridiales (guild)	Depleted	Linked to anti-inflammatory effects; its reduction indicates loss of protective microbial functions.
Prevotella amnii	Enriched	Associated with inflammation and estrogen metabolism, particularly in postmenopausal breast cancer.
Lactobacillus vaginalis	Depleted	Supports mucosal health and immune balance; depletion is a marker of dysbiosis, especially postmenopause.

Interventions

The validation process aligns microbiome-targeted interventions (MBTIs) with microbial dysbiosis correction and clinical improvements. Breast cancer treatments may benefit from incorporating strategies to restore microbial balance, such as dietary SCFA supplementation, probiotics, and precision microbiome therapies.

Class	Interventions	Mechanisms of Action	MBTI Status
Pharmaceutical	Tamoxifen	Selectively binds estrogen receptors (ER), preventing estrogen from binding to the receptor, inhibiting cancer cell proliferation. It also downregulates estrogen-dependent gene expression, preventing tumor growth in estrogen receptor-positive breast cancers.[45][46]	Validated
Non-Pharmaceutical	Exercise Therapy	Increases circulation, reduces systemic inflammation, enhances immune function, and helps manage cancer-related fatigue by promoting the release of endorphins.[47]	Promising
	Fecal Microbiota Transplantation (FMT)	FMT directly targets the gut microbiome by reintroducing a healthy microbiota, thus potentially altering microbial-related immune pathways. It has been shown to improve chemotherapy efficacy by restoring a more favorable microbial composition that supports anti-tumor immune responses.[48][49]	Experimental
Supplements	Vitamin D	As vitamin D downregulates ERα expression via inhibition of NF-κB, it may increase the sensitivity to tamoxifen through induction of functional ERα in ER-negative cancer cells. Vitamin D combined with tamoxifen may be effective in tamoxifen-resistant tumors, and its concurrent use with aromatase inhibitors may be another suitable therapeutic option.[50][51][52]	Validated
	Short-Chain Fatty Acid (SCFA)	SCFAs play a crucial role in maintaining gut health by promoting gut barrier integrity, regulating immune responses, and reducing systemic inflammation. SCFAs, especially butyrate, are known to inhibit histone deacetylases (HDACs), leading to changes in gene expression that promote apoptosis (programmed cell death) and inhibit tumor cell proliferation.[53][54]	Under Investigation
	Flavonoids	Flavonoids, including apigenin, quercetin, and genistein, have been shown to inhibit estrogen receptor signaling, suppress aromatase activity, and reduce estrogen synthesis within cancer cells. They also induce apoptosis and inhibit cell proliferation.[55]	Promising Candidate

Class	Interventions	Mechanisms of Action	MBTI Status
Nutritional Therapy	Probiotics	Probiotics produce short-chain fatty acids (SCFAs) that support gut health, enhance immune function, and help in reducing systemic inflammation. These actions can influence cancer progression, as inflammation and immune dysregulation are often linked to tumor development.[56][57][x][58]	Promising Candidate
	High-Fiber Diet	Regulates blood sugar and insulin levels, which affect breast cancer risk and prognosis. Fiber promotes the growth of beneficial gut bacteria like *Bifidobacterium* and *Lactobacillus*.[59][x]	Validated
Drug Repurposing	Aspirin	Inhibits COX-2, reducing inflammation and tumor growth by suppressing pro-inflammatory cytokines.[60][61]	Promising Candidate
	Metformin	Regulates insulin levels, inhibits mTOR, and reduces cell proliferation in cancer cells.[62]	Promising Candidate

STOPs

The STOPs initiative advocates for the careful re-evaluation of common clinical practices that could potentially disrupt the microbiome and increase breast cancer risk. It emphasizes the significant role the microbiome plays in modulating immune responses, systemic inflammation, and estrogen metabolism. Recent studies suggest that specific practices, such as the routine use of chlorhexidine mouthwash, prolonged estrogen supplementation, and increased use of antibiotics, can negatively impact the microbiome, contributing to dysbiosis (microbial imbalance) that might promote breast cancer progression, particularly hormone receptor-positive and chemoresistant breast cancers.[63][x][64] Chlorhexidine is widely used for oral hygiene but has been found to alter the oral microbiome, reducing microbial diversity and promoting the overgrowth of potentially harmful species like Proteobacteria.[65] These shifts in the oral microbiome are linked to systemic inflammation and estrogen metabolism alterations, which could exacerbate hormone-driven breast cancer.

STOP	Microbiome Impact	Breast Cancer Risk
Chlorhexidine Mouthwash	Reduces oral microbial diversity, increases Proteobacteria, decreases beneficial microbes like Fusobacterium and Porphyromonas.	Potential causal link to breast cancer through oral and gut dysbiosis, influencing estrogen metabolism and immune function.

References

1. Current and future burden of breast cancer: Global statistics for 2020 and 2040.. Arnold M, Morgan E, Rumgay H, Mafra A, Singh D, Laversanne M, Vignat J, Gralow JR, Cardoso F, Siesling S, Soerjomataram I.. (Breast. 2022 Dec;66:15-23.)

2. Breast cancer but not the menopausal status is associated with small changes of the gut microbiota. Zeber-Lubecka N, Kulecka M, Jagiełło-Gruszfeld A, Dąbrowska M, Kluska A, Piątkowska M, Bagińska K, Głowienka M, Surynt P, Tenderenda M, Mikula M, Ostrowski J.. (Front Oncol. 2024 Jan 24;14:1279132)
3. Breast cancer but not the menopausal status is associated with small changes of the gut microbiota. Zeber-Lubecka N, Kulecka M, Jagiełło-Gruszfeld A, Dąbrowska M, Kluska A, Piątkowska M, Bagińska K, Głowienka M, Surynt P, Tenderenda M, Mikula M, Ostrowski J.. (Front Oncol. 2024 Jan 24;14:1279132)
4. BRCA Genes: The Role in Genome Stability, Cancer Stemness and Therapy Resistance. Gorodetska I, Kozeretska I, Dubrovska A.. (J Cancer 2019; 10(9):2109-2127.)
5. A comprehensive analysis of breast cancer microbiota and host gene expression. Thompson KJ, Ingle JN, Tang X, Chia N, Jeraldo PR, Walther-Antonio MR, Kandimalla KK, Johnson S, Yao JZ, Harrington SC, Suman VJ, Wang L, Weinshilboum RL, Boughey JC, Kocher JP, Nelson H, Goetz MP, Kalari KR.. (PLoS One. 2017 Nov 30;12(11):e0188873.)
6. Antibiotic use in relation to the risk of breast cancer. Velicer CM, Heckbert SR, Lampe JW, Potter JD, Robertson CA, Taplin SH.. (JAMA. 2004 Feb 18;291(7):827-35.)
7. Effects of chlorhexidine mouthwash on the oral microbiome. Brookes ZLS, Belfield LA, Ashworth A, Casas-Agustench P, Raja M, Pollard AJ, Bescos R.. (J Dent. 2021 Oct;113:103768)
8. Diabetes as a Risk Factor for Breast Cancer. Eketunde AO.. (Cureus. 2020 May 7;12(5):e8010)
9. Diabetes mellitus and risk of breast cancer: a large-scale, prospective, population-based study. Xiong, F., Wang, J., Nierenberg, J.L. et al.. (Br J Cancer 129, 648–655 (2023))
10. History of benign breast disease and risk of breast cancer among women in China: a case-control study. Dorjgochoo T, Deming SL, Gao YT, Lu W, Zheng Y, Ruan Z, Zheng W, Shu XO.. (Cancer Causes Control. 2008 Oct;19(8):819-28)
11. Breast cancer patients from the Midwest region of the United States have reduced levels of short-chain fatty acid-producing gut bacteria.. Shrode RL, Knobbe JE, Cady N, Yadav M, Hoang J, Cherwin C, Curry M, Garje R, Vikas P, Sugg S, Phadke S, Filardo E, Mangalam AK.. (Sci Rep. 2023)
12. A comprehensive analysis of breast cancer microbiota and host gene expression. Thompson KJ, Ingle JN, Tang X, Chia N, Jeraldo PR, Walther-Antonio MR, Kandimalla KK, Johnson S, Yao JZ, Harrington SC, Suman VJ, Wang L, Weinshilboum RL, Boughey JC, Kocher JP, Nelson H, Goetz MP, Kalari KR.. (PLoS One. 2017 Nov 30;12(11):e0188873.)
13. Breast cancer patients from the Midwest region of the United States have reduced levels of short-chain fatty acid-producing gut bacteria.. Shrode RL, Knobbe JE, Cady N, Yadav M, Hoang J, Cherwin C, Curry M, Garje R, Vikas P, Sugg S, Phadke S, Filardo E, Mangalam AK.. (Sci Rep. 2023)
14. The Unknown Effect of Antibiotic-Induced Dysbiosis on the Gut Microbiota. Aleksandr Birg, Nathaniel L. Ritz, Henry C. Lin. (Microbiome and Metabolome in Diagnosis, Therapy, and other Strategic Applications, Academic Press, 2019,)
15. Breast cancer in postmenopausal women is associated with an altered gut metagenome.. Zhu J, Liao M, Yao Z, Liang W, Li Q, Liu J, Yang H, Ji Y, Wei W, Tan A, Liang S, Chen Y, Lin H, Zhu X, Huang S, Tian J, Tang R, Wang Q, Mo Z.. (Microbiome. 2018)
16. Antibiotic use in relation to the risk of breast cancer.. Velicer CM, Heckbert SR, Lampe JW, Potter JD, Robertson CA, Taplin SH.. (JAMA. 2004)
17. Comprehensive profiles and diagnostic value of menopausal-specific gut microbiota in premenopausal breast cancer.. Hou MF, Ou-Yang F, Li CL, Chen FM, Chuang CH, Kan JY, Wu CC, Shih SL, Shiau JP, Kao LC, Kao CN, Lee YC, Moi SH, Yeh YT, Cheng CJ, Chiang CP.. (Exp Mol Med. 2021)
18. Characterization of the microbiome of nipple aspirate fluid of breast cancer survivors.. Chan AA, Bashir M, Rivas MN, Duvall K, Sieling PA, Pieber TR, et al.. (Sci Rep. 2016)
19. The role of cadmium and nickel in estrogen receptor signaling and breast cancer: metalloestrogens or not?. Aquino NB, Sevigny MB, Sabangan J, Louie MC.. (J Environ Sci Health C Environ Carcinog Ecotoxicol Rev. 2012;30(3):189-224)
20. Impact of heavy metals on breast cancer (Review). Ali, A.S., Nazar, M.E., Mustafa, R.M., Hussein, S., Qurbani, K., & Ahmed, S.K. (2024).. (World Academy of Sciences Journal, 6, 4.)
21. Cadmium and breast cancer – Current state and research gaps in the underlying mechanisms. Tarhonska, K., Lesicka, M., Janasik, B., Roszak, J., Reszka, E., Braun, M., Kołacińska-Wow, A., & Jabłońska, E. (2022). (Toxicology Letters, 361, 29-42)
22. Zinc Deficiency as a General Feature of Cancer: a Review of the Literature. Sugimoto, R., Lee, L., Tanaka, Y. et al.. (Biol Trace Elem Res 202, 1937–1947 (2024))

23. The role of cadmium and nickel in estrogen receptor signaling and breast cancer: metalloestrogens or not?. Aquino NB, Sevigny MB, Sabangan J, Louie MC.. (J Environ Sci Health C Environ Carcinog Ecotoxicol Rev. 2012;30(3):189-224)

24. Impact of heavy metals on breast cancer (Review). Ali, A.S., Nazar, M.E., Mustafa, R.M., Hussein, S., Qurbani, K., & Ahmed, S.K. (2024).. (World Academy of Sciences Journal, 6, 4.)

25. Relationships Between Biological Heavy Metals and Breast Cancer: A Systematic Review and Meta-Analysis. Liu, L., Chen, J., Liu, C., Luo, Y., Chen, J., Fu, Y., Xu, Y., Wu, H., Li, X., & Wang, H. (2022). (Frontiers in Nutrition, 9, 838762.)

26. Cadmium and breast cancer – Current state and research gaps in the underlying mechanisms. Tarhonska, K., Lesicka, M., Janasik, B., Roszak, J., Reszka, E., Braun, M., Kołacińska-Wow, A., & Jabłońska, E. (2022). (Toxicology Letters, 361, 29-42)

27. The role of cadmium and nickel in estrogen receptor signaling and breast cancer: metalloestrogens or not?. Aquino NB, Sevigny MB, Sabangan J, Louie MC.. (J Environ Sci Health C Environ Carcinog Ecotoxicol Rev. 2012;30(3):189-224)

28. Cadmium exposure and risk of breast cancer: A meta-analysis. Florez-Garcia VA, Guevara-Romero EC, Hawkins MM, Bautista LE, Jenson TE, Yu J, Kalkbrenner AE.. (Environ Res. 2023 Feb 15;219:115109)

29. Impact of heavy metals on breast cancer (Review). Ali, A.S., Nazar, M.E., Mustafa, R.M., Hussein, S., Qurbani, K., & Ahmed, S.K. (2024).. (World Academy of Sciences Journal, 6, 4.)

30. Zinc Deficiency as a General Feature of Cancer: a Review of the Literature. Sugimoto, R., Lee, L., Tanaka, Y. et al.. (Biol Trace Elem Res 202, 1937–1947 (2024))

31. Relationships Between Biological Heavy Metals and Breast Cancer: A Systematic Review and Meta-Analysis. Liu, L., Chen, J., Liu, C., Luo, Y., Chen, J., Fu, Y., Xu, Y., Wu, H., Li, X., & Wang, H. (2022). (Frontiers in Nutrition, 9, 838762.)

32. Zinc Deficiency as a General Feature of Cancer: a Review of the Literature. Sugimoto, R., Lee, L., Tanaka, Y. et al.. (Biol Trace Elem Res 202, 1937–1947 (2024))

33. Impact of heavy metals on breast cancer (Review). Ali, A.S., Nazar, M.E., Mustafa, R.M., Hussein, S., Qurbani, K., & Ahmed, S.K. (2024).. (World Academy of Sciences Journal, 6, 4.)

34. Relationships Between Biological Heavy Metals and Breast Cancer: A Systematic Review and Meta-Analysis. Liu, L., Chen, J., Liu, C., Luo, Y., Chen, J., Fu, Y., Xu, Y., Wu, H., Li, X., & Wang, H. (2022). (Frontiers in Nutrition, 9, 838762.)

35. Relationships Between Biological Heavy Metals and Breast Cancer: A Systematic Review and Meta-Analysis. Liu, L., Chen, J., Liu, C., Luo, Y., Chen, J., Fu, Y., Xu, Y., Wu, H., Li, X., & Wang, H. (2022). (Frontiers in Nutrition, 9, 838762.)

36. Impact of heavy metals on breast cancer (Review). Ali, A.S., Nazar, M.E., Mustafa, R.M., Hussein, S., Qurbani, K., & Ahmed, S.K. (2024).. (World Academy of Sciences Journal, 6, 4.)

37. Breast cancer patients from the Midwest region of the United States have reduced levels of short-chain fatty acid-producing gut bacteria.. Shrode RL, Knobbe JE, Cady N, Yadav M, Hoang J, Cherwin C, Curry M, Garje R, Vikas P, Sugg S, Phadke S, Filardo E, Mangalam AK.. (Sci Rep. 2023)

38. Breast cancer in postmenopausal women is associated with an altered gut metagenome. Zhu, J., Liao, M., Yao, Z. et al.. (Microbiome 6, 136 (2018))

39. Breast cancer in postmenopausal women is associated with an altered gut metagenome. Zhu, J., Liao, M., Yao, Z. et al.. (Microbiome 6, 136 (2018))

40. Breast cancer but not the menopausal status is associated with small changes of the gut microbiota. Zeber-Lubecka N, Kulecka M, Jagiełło-Gruszfeld A, Dąbrowska M, Kluska A, Piątkowska M, Bagińska K, Głowienka M, Surynt P, Tenderenda M, Mikula M, Ostrowski J.. (Front Oncol. 2024 Jan 24;14:1279132)

41. Comprehensive profiles and diagnostic value of menopausal-specific gut microbiota in premenopausal breast cancer. Hou, MF., Ou-Yang, F., Li, CL. et al.. (Exp Mol Med 53, 1636–1646 (2021).)

42. Microbial dysbiosis is associated with human breast cancer.. Xuan C, Shamonki JM, Chung A, Dinome ML, Chung M, Sieling PA, Lee DJ.. (PLoS One. 2014 Jan 8;9(1):e83744)

43. The oral microbiome and breast cancer and nonmalignant breast disease, and its relationship with the fecal microbiome in the Ghana Breast Health Study. Wu Z, Byrd DA, Wan Y, Ansong D, Clegg-Lamptey JN, Wiafe-Addai B, Edusei L, Adjei E, Titiloye N, Dedey F, Aitpillah F, Oppong J, Vanderpuye V, Osei-Bonsu E, Dagnall CL, Jones K, Hutchinson A, Hicks BD, Ahearn TU, Shi J, Knight R, Biritwum R, Yarney J, Wiafe S, Awuah B, Nyarko K, Figueroa JD, Sinha R, Garcia-Closas M, Brinton LA, Vogtmann E.. (Int J Cancer. 2022 Oct 15;151(8):1248-1260)

44. Gut microbiome associations with breast cancer risk factors and tumor characteristics: a pilot study. Wu AH, Tseng C, Vigen C, Yu Y, Cozen W, Garcia AA, Spicer D.. (Breast Cancer Res Treat. 2020 Jul;182(2):451-463)

45. Interaction between Estrogen Receptors and p53: A Broader Role for Tamoxifen?. Gokul M Das, Chetan C Oturkar, Vishnu Menon. (Endocrinology, Volume 166, Issue 3, March 2025, bqaf020)

46. The gut microbiota during tamoxifen therapy in patients with breast cancer. Hillege, L. E., Barnett, D. J., Ziemons, J., Aarnoutse, R., Van Geel, R., De Boer, M., Van Riet, Y. E., Vincent, J., Penders, J., & Smidt, M. L. (2025). (Scientific Reports, 15(1), 1-12)

47. Exercise, Gut Microbiome, and Gastrointestinal Diseases: Therapeutic Impact and Molecular Mechanisms. Hawley, J. A., Forster, S. C., & Giles, E. M. (2025). (Gastroenterology)

48. Fecal microbiota transplantation in cancer management: Current status and perspectives. Chen, D., Wu, J., Jin, D., Wang, B., & Cao, H. (2018). (International Journal of Cancer, 145(8), 2021)

49. Antitumor effects of fecal microbiota transplantation: Implications for microbiome modulation in cancer treatment. Xu, H., Cao, C., Ren, Y., Weng, S., Liu, L., Guo, C., Wang, L., Han, X., Ren, J., & Liu, Z. (2022). (Frontiers in Immunology, 13, 949490.)

50. Vitamin D: An essential adjuvant therapeutic agent in breast cancer. Thabet, R. H., Gomaa, A. A., Matalqah, L. M., & Shalaby, E. M. (2022). (The Journal of International Medical Research, 50(7), 03000605221113800)

51. Repurposing vitamin D for treatment of human malignancies via targeting tumor microenvironment. Wu, X., Hu, W., Lu, L., Zhao, Y., Zhou, Y., Xiao, Z., Zhang, L., Zhang, H., Li, X., Li, W., Wang, S., Cho, C. H., Shen, J., & Li, M. (2018). (Acta Pharmaceutica Sinica. B, 9(2), 203)

52. Randomized control trial of moderate dose vitamin D alters microbiota stability and metabolite networks in healthy adults. Wyatt, M., Choudhury, A., Dohlen, G. V., Heileson, J. L., Forsse, J. S., Rajakaruna, S., Zec, M., Tfaily, M. M., & Greathouse, L. (2024). (Microbiology Spectrum, 12(10), e00083-24)

53. Short-chain fatty acids in cancer pathogenesis.. Feitelson, M.A., Arzumanyan, A., Medhat, A. et al.. (Cancer Metastasis Rev 42, 677–698 (2023))

54. Potential antitumor effects of short-chain fatty acids in breast cancer models. Muradás TC, Freitas RD, Gonçalves JI, Xavier FA, Marinowic DR.. (Am J Cancer Res. 2024 May 15;14(5):1999-2019)

55. Natural products and derivatives for breast cancer treatment: From drug discovery to molecular mechanism.. Zhang, J., Wu, Y., Li, Y., Li, S., Liu, J., Yang, X., Xia, G., & Wang, G. (2024). (Phytomedicine, 129, 155600)

56. Effect of Probiotics in Breast Cancer: A Systematic Review and Meta-Analysis. Thu, M. S., Ondee, T., Nopsopon, T., K Farzana, I. A., Fothergill, J. L., Hirankarn, N., Campbell, B. J., & Pongpirul, K. (2023). (Biology, 12(2), 280)

57. Potential effect of probiotics in the treatment of breast cancer. Mendoza, L. (2019). (Oncology Reviews, 13(2), 422)

58. Microbial Therapy and Breast Cancer Management: Exploring Mechanisms, Clinical Efficacy, and Integration within the One Health Approach. Filippou, C., Themistocleous, S. C., Marangos, G., Panayiotou, Y., Fyrilla, M., Kousparou, C. A., Pana, D., Tsioutis, C., Johnson, E. O., & Yiallouris, A. (2024). (International Journal of Molecular Sciences, 25(2), 1110)

59. Dietary fiber consumption and outcomes of different cancers: an umbrella review. He X., Hou J., Liu L., Chen X., Zhang L., Pang C., Tong Y., Li H., Chen F., Peng R., & Shi Z. (2025).. (Food & Nutrition Research, 69.)

60. Aspirin suppresses breast cancer metastasis to lung by targeting anoikis resistance. Xu R, Yan Y, Zheng X, Zhang H, Chen W, Li H, Dong Z.. (Carcinogenesis. 2022 Mar 24;43(2):104-114)

61. Aspirin Use and Survival Among Patients With Breast Cancer: A Systematic Review and Meta-Analysis. Baker, A., & Kartsonaki, C. (2024). (The Oncologist, 29(1), e1-e14.)

62. Metformin and Cancer in Type 2 Diabetes.. Park HK.. (Diabetes Metab J. 2013;37(2):113-116.)

63. Antibiotic use in relation to the risk of breast cancer. Velicer CM, Heckbert SR, Lampe JW, Potter JD, Robertson CA, Taplin SH.. (JAMA. 2004 Feb 18;291(7):827-35)

64. Mouthwash Effects on the Oral Microbiome: Are They Good, Bad, or Balanced?. Brookes Z, Teoh L, Cieplik F, Kumar P.. (Int Dent J. 2023 Nov;73 Suppl 2(Suppl 2):S74-S81)

65. Effects of Chlorhexidine mouthwash on the oral microbiome. Bescos, R., Ashworth, A., Cutler, C. et al.. (Sci Rep 10, 5254 (2020).)

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